Abhishek Singh

CS undergrad at KIIT building end-to-end ML systems — from EDA and model training to deployment. Experienced with Python, Scikit-learn, TensorFlow, and FastAPI.

Seeking Data Science / Data Analytics internships

gmail | portfolio | linkedin | github | itch.io

Education

Kalinga Institute of Industrial Technology (KIIT) — Bhubaneswar, India
Bachelor of Technology in Computer Science and Engineering — *2023 – Present | CGPA: 8.48*

Projects

Talk with CSV | *React, FastAPI, Groq AI (Llama 3.3), Python, Pandas, Matplotlib* | Live | GitHub — 2026

- Built a full-stack AI-powered analytics platform enabling natural-language queries over uploaded CSV datasets
- Designed a FastAPI backend using Groq LLM APIs to generate insights, tables, and visualization instructions
- Implemented automated chart generation and deployed the application using React, Render, and modern REST APIs

RAG + Persona Engine | *Python, Ollama, ChromaDB, Sentence-Transformers, Streamlit* | Live | GitHub — 2026

- Built a fully local RAG pipeline that segments long conversation logs using embedding-drift topic detection and fixed-interval checkpointing, summarizing each via a local LLM
- Designed dual-collection retrieval in ChromaDB combining topic summaries with raw message chunks for grounded, context-aware Q&A
- Engineered an evidence-grounded persona extraction module and deployed an interactive Streamlit chatbot, running entirely on local inference with no paid external APIs

Disease Risk Predictor | *Python, Scikit-learn, Streamlit, Docker* | Live | GitHub — 2026

- Built an end-to-end ML web app predicting Diabetes and Heart Disease risk from real medical datasets
- Applied SMOTE for class imbalance and GridSearchCV tuning; achieved 92.44% accuracy on heart disease
- Containerized with Docker and deployed live on Render

Network Intrusion Detection System | *Python, Scikit-learn, TensorFlow, Pandas, NumPy* | GitHub — 2026

- Developed an anomaly-based intrusion detection system using the KDD Cup 1999 cybersecurity dataset
- Implemented and compared Isolation Forest and Autoencoder models for unsupervised attack detection
- Achieved **95.81% accuracy**, 98.37% recall, and 91.87% F1-score, significantly outperforming Isolation Forest

Game Development – IIT Guwahati Game Jam (GameDeon) | *Godot, GDScript* | itch.io — 2025

- Secured **7th position among 50 teams** nationwide; invited to IIT Guwahati for final showcase

Technical Skills

Languages: Python, SQL, R, Java, C, C++, JavaScript
Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow, EDA, Feature Engineering, Model Evaluation, SMOTE, Hyperparameter Tuning
AI / LLM Engineering: RAG Pipelines, Ollama, ChromaDB, Sentence-Transformers, Embeddings, Prompt Engineering, Groq API
Algorithms: Random Forest, Gradient Boosting, Logistic Regression, SVM, Isolation Forest, Autoencoders, TF-IDF, Cosine Similarity
Visualization: Matplotlib, Seaborn
Web & APIs: React, FastAPI, REST APIs, HTML, CSS
Databases & Tools: SQL, MariaDB, SQLite, Git, GitHub, Docker, Linux, Jupyter Notebook, Streamlit

Certifications

IBM Data Science Professional Certificate — Coursera
Tools for Data Science, Data Science Methodology, Python for Data Science, AI & Machine Learning (In Progress)

Google Data Analytics Certificate — Coursera
Data cleaning, analysis, and visualization using spreadsheets, SQL, R, and Tableau

Additional

Competitive Programming
Solved 250+ problems on LeetCode and CodeChef using Python and C++

GameJam Hackathon
Participated in the GameDeon Game Jam organized by IIT Guwahati